LUMEXA
Space-themed coding education for K–12

AGES 12–18 | MOST POPULAR

CURRICULUM OVERVIEW

Advanced Data Science

Build 15+ Real AI & Data Science Projects

A complete, project-driven journey from your first line of Python to arcade games, data dashboards and machine learning models you can deploy. Students don't just learn about AI and data. They write the code, build the projects and prove it with tests.

15+ REAL AI & DATA SCIENCE PROJECTS
Portfolio-ready work

6 TESTS TO PROVE IT
5 course tests + 1 final test
on the whole curriculum

5	40	46	12–18
Courses	Lessons	Sessions in total	Age range

LEARNING PROGRESSION

01 — Python for Young Builders
From block coding to real Python. Build 3 apps and a mini game.

02 — Python Arcade Games
Pygame, sprites and collisions. Ship 3 playable arcade games.

03 — Python for Data
Pandas, data cleaning and statistics on real datasets.

04 — Data Visualisation
Charts, Plotly and Dash dashboards that tell a story.

05 — Machine Learning Projects
Train, tune and deploy models that predict and recommend.

Learn by building, not by watching
Every lesson pairs a real concept with real, runnable code. No shallow demos, no black boxes.

A portfolio, not just a certificate
Students finish with 15+ working projects they can show, explain and keep building on.

Tested, so progress is real
Five course tests and one final test on the whole curriculum prove what students can do.

COURSE 01 · LESSONS 1–8

3 REAL PROJECTS | 1 COURSE TEST

Python for Young Builders

Go from block coding to real Python by building your first apps and games

Python 3 | Variables & Input | Loops & Lists | Mini Games

OVERVIEW

The foundation of everything. Your child moves from blocks to real Python: variables, decisions, loops and lists, and finishes with a mini game of their own. No fluff, just real code.

PORTFOLIO PROJECTS

- **Personal Quiz App**
- **Simple Calculator**
- **Word Guessing Game**

WHY IT MATTERS

Every program, from a quiz app to an AI model, is built from these same building blocks. Students learn to think in code, break problems down and ship small projects that actually run.

COURSE TEST

Python for Young Builders — Course Test

Show what you have built and learned before moving on to the next course.

LESSON BREAKDOWN

1. **From Blocks to Words — Hello, Python!**
   Set up your environment and preview the module
2. **Variables and User Input**
   Part 1 · Personal Quiz App
3. **Printing and String Formatting**
   Part 2 · Personal Quiz App
4. **If/Else Decisions**
   Part 1 · Simple Calculator
5. **While Loops and For Loops**
   Part 2 · Simple Calculator
6. **Lists and Simple Data**
   Part 1 · Word Guessing Game
7. **Building a Mini Quiz App**
   Part 2 · Word Guessing Game
8. **Final Project — Our Own Mini Game!**
   Capstone · combine all 3 projects

COURSE 02 · LESSONS 1–8

3 REAL PROJECTS | 1 COURSE TEST

Python Arcade Games

Build playable arcade games with Pygame and share them with the world

Python | Pygame | Sprites & Sound | Game Loop

OVERVIEW

Your child builds real arcade games: a space shooter, a maze runner and a brick breaker. They learn how game loops, collisions, enemies and scoring really work, then package and share their games.

PORTFOLIO PROJECTS

- **Space Shooter**
- **Maze Runner with Timer**
- **Brick Breaker Clone**

WHY IT MATTERS

Games are the most motivating way to master programming. Students practise logic, maths and problem-solving every time they tweak a level, and finish with games friends and family can play.

COURSE TEST

Python Arcade Games — Course Test

Show what you have built and learned before moving on to the next course.

LESSON BREAKDOWN

1. **Python Basics for Game Development**
   Set up your environment and preview the module
2. **Pygame Setup and the Game Loop**
   Part 1 · Space Shooter
3. **Sprites, Images, and Movement**
   Part 2 · Space Shooter
4. **Collision Detection**
   Part 1 · Maze Runner with Timer
5. **Enemies, AI Behavior, and Scoring**
   Part 2 · Maze Runner with Timer
6. **Sound Effects and Music**
   Part 1 · Brick Breaker Clone
7. **Game States — Menu, Play, Game Over**
   Part 2 · Brick Breaker Clone
8. **Packaging and Sharing Your Game**
   Capstone · polish and share all 3 games

COURSE 03 · LESSONS 1–8

3 REAL PROJECTS | 1 COURSE TEST

Python for Data

Turn raw data into answers with Pandas and real-world datasets

Python | Pandas | Jupyter Notebooks | CSV & Excel

OVERVIEW

Your child learns how data scientists work: loading real data, cleaning it, grouping it and finding patterns. They analyse sports, school and city datasets, the same skills used in business and research.

PORTFOLIO PROJECTS

- **Sports Performance Analysis**
- **School Grade Analysis**
- **City Population Trends**

WHY IT MATTERS

Data drives decisions in every industry. Students learn the full analysis workflow, from messy files to clear insights, which is the first step of every data science and AI project.

COURSE TEST

Python for Data — Course Test

Show what you have built and learned before moving on to the next course.

LESSON BREAKDOWN

1. **Why Data Science Matters (Real Examples)**
   Set up your environment and preview the module
2. **Python Refresher and Jupyter Setup**
   Part 1 · Sports Performance Analysis
3. **Loading Data from CSV and Excel**
   Part 2 · Sports Performance Analysis
4. **Pandas — Selecting, Filtering, Sorting**
   Part 1 · School Grade Analysis
5. **Cleaning Data — Nulls, Duplicates, Types**
   Part 2 · School Grade Analysis
6. **Aggregations — Group By, Pivot Tables**
   Part 1 · City Population Trends
7. **Statistical Summaries and Correlation**
   Part 2 · City Population Trends
8. **Full Analysis Project with a Real Dataset**
   Capstone · polish and present all 3 analyses

COURSE 04 · LESSONS 1–8

3 REAL PROJECTS | 1 COURSE TEST

Data Visualisation

Create charts and interactive dashboards that tell a clear story

Matplotlib | Seaborn | Plotly | Dash

OVERVIEW

Your child learns to make data visible: from classic charts to interactive dashboards. They build a climate dashboard, a market trends chart and a sports comparison tool that anyone can explore.

PORTFOLIO PROJECTS

- **Climate Change Dashboard**
- **Market Trends Chart**
- **Sports Comparison Visualiser**

WHY IT MATTERS

A good chart can change minds. Students learn which chart fits which data, how colour and labels shape the message, and how to present findings to people who are not technical.

COURSE TEST

Data Visualisation — Course Test

Show what you have built and learned before moving on to the next course.

LESSON BREAKDOWN

1. **The Science of Good Data Visualisation**
   Set up your environment and preview the module
2. **Matplotlib: Line, Bar, Scatter Plots**
   Part 1 · Climate Change Dashboard
3. **Seaborn: Statistical Visualisations**
   Part 2 · Climate Change Dashboard
4. **Choosing the Right Chart for Your Data**
   Part 1 · Market Trends Chart
5. **Color, Labels, and Storytelling**
   Part 2 · Market Trends Chart
6. **Interactive Charts with Plotly**
   Part 1 · Sports Comparison Visualiser
7. **Building a Dash Dashboard**
   Part 2 · Sports Comparison Visualiser
8. **Your Own Interactive Dashboard**
   Capstone · polish and publish all 3 projects

COURSE 05 · LESSONS 1–8

3 REAL PROJECTS | 1 COURSE TEST

Machine Learning Projects

Train, test and deploy models that predict, classify and recommend

Python | scikit-learn | Pandas | Model APIs

OVERVIEW

Your child learns how machine learning really works, with no maths jargon. They train models that predict house prices, spot customers about to leave and recommend movies, then deploy one as an API.

PORTFOLIO PROJECTS

- **House Price Predictor**
- **Customer Churn Predictor**
- **Movie Recommendation Engine**

WHY IT MATTERS

Every ML system, from spam filters to recommendation engines, follows the same workflow: prepare data, train a model, test it honestly and explain the result. This course puts it all together.

COURSE TEST

Machine Learning Projects — Course Test

Show what you have built and learned before moving on to the next course.

LESSON BREAKDOWN

1. **How Machine Learning Works (No Maths Jargon)**
   Set up your environment and preview the module
2. **Types of ML — Regression vs Classification**
   Part 1 · House Price Predictor
3. **Train/Test Splits and Model Evaluation**
   Part 2 · House Price Predictor
4. **Linear Regression From Scratch**
   Part 1 · Customer Churn Predictor
5. **Decision Trees and Random Forests**
   Part 2 · Customer Churn Predictor
6. **Feature Engineering and Selection**
   Part 1 · Movie Recommendation Engine
7. **Model Tuning and Improving Accuracy**
   Part 2 · Movie Recommendation Engine
8. **Deploying a Prediction Model as an API**
   Capstone · polish and ship all 3 models

The Complete Journey

Across 5 courses, 40 lessons and 6 tests, students progress from their first Python program to building, visualising and deploying real AI and data science projects, finishing with 15+ working projects for their portfolio.

5 COURSES · 15+ PROJECTS

01 — Python for Young Builders
Personal Quiz App · Simple Calculator · Word Guessing Game
8 lessons + test

02 — Python Arcade Games
Space Shooter · Maze Runner with Timer · Brick Breaker Clone
8 lessons + test

03 — Python for Data
Sports Performance Analysis · School Grade Analysis · City Population Trends
8 lessons + test

04 — Data Visualisation
Climate Change Dashboard · Market Trends Chart · Sports Comparison Visualiser
8 lessons + test

05 — Machine Learning Projects
House Price Predictor · Customer Churn Predictor · Movie Recommendation Engine
8 lessons + test

HOW STUDENTS ARE TESTED · 6 TESTS

- COURSE TEST 1 — Python for Young Builders
- COURSE TEST 2 — Python Arcade Games
- COURSE TEST 3 — Python for Data
- COURSE TEST 4 — Data Visualisation
- COURSE TEST 5 — Machine Learning Projects

FINAL TEST — One final test on the whole curriculum
Covers Python, games, data, visualisation and machine learning

BY THE END OF THE PATH, STUDENTS CAN

- Write real Python programs and build playable games from scratch.
- Clean, analyse and visualise real data with Pandas, Matplotlib and Plotly.
- Train, test and deploy machine learning models as a working API.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.